

Market Survey

How Far Will 3D PRINTING Go



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Three-dimensional (3D) printing can help industry think faster, innovate better and broaden the horizons of the manufacturing sector. But how far will the 3D printing industry grow in India? What are the growing trends in this sector? What are the growth drivers? While trying to figure out the answers to these questions, we got some insights from stake holders in this industry, which we share with our readers in this article.

3D printing is an example of technology and manufacturing working together to become more flexible and scalable—allowing for enhanced efficiency and quality. 3D printer software enables users to create a digital model from which physical objects can be made. The primary benefit of this technology is the ability to manufacture customised parts on ‘as needed’ basis without costly machinery or dies for casting.

3D printing comes with several advantages like lower resource requirement, faster production cycles, flexible designs and substantial savings on tooling compared to traditional manufacturing technologies such as injection moulding, computer numerical control (CNC) machining and vacuum casting. This is very conducive for making low-volume customised products of high value.

3D printing technology helps industries streamline their design processes and lower production costs. Today, manufacturers

across form a broad spectrum of industries including automotive, aerospace, dental, discrete, high-tech and medical products. All these industries are actively piloting and using 3D printing technologies. Prototyping continues to be the dominant reason why enterprises adopt 3D printing, with the opportunity of improving new product development and time-to-market being long-term goals.

Considering the growing importance of 3D printing in the electronics manufacturing ecosystem, we decided to take a look at the 3D printing industry in India to understand how it has been shaping up. In this edition, however, rather than merely presenting qualitative perspectives shared by industry members, we also thought of sharing industry trends that we arrived at through a mini-survey involving a sample group of senior-level industry members.

Growth prospects

3D printing is emerging as the fastest growing printing technology worldwide. In India, the market for 3D printers is at its nascent stage. However, it offers huge growth opportunities in the coming years. According to 6Wresearch, a market intelligence and advisory company, the 3D printer market in India is projected to touch US\$ 79 million by 2021.

In line with broad industry consensus, more than 90 per cent of survey participants stated that the industry is poised to grow during FY 2016-17. However, they differed with respect to the expected growth percentage. More than half of the survey participants expect the industry to grow more than 15 per cent (Fig. 1).

According to the survey participants, the thrust on domestic production, demand for lean manufacturing and increasing penetration across various applications, coupled with the Make in India campaign, will spur the 3D printer market in India. With initiatives like Make in India, the domestic manufacturing sector is being encouraged, which automatically will play a pivotal role in the growth of the local 3D printing industry.

Methodology

For the survey, 42 senior professionals involved in the 3D printing industry were randomly selected. Seventeen of them were available for sharing their inputs. This sample is a microcosmic representation of India's 3D printing sector. Survey participants were requested to share their insight on:

1. The growth of the Indian 3D printing industry in FY 2016-17
2. Expected growth per cent
3. The major demand-generating applications
4. Customer categorisation
5. Hindrances that impact the growth of this sector
6. Technology trends in this sector

The responses obtained from the interviews were then collated and analysed.